

Introduction

A common treatment process to protect aluminum from physical and chemical damage is anodization, a process that can increase the thickness of the aluminum oxide layer already present on aluminum. This is done by placing it in an acidic electrolyte solution and applying a voltage, with aluminum piece being the anode in the circuit, forcing a reaction that builds upon the oxide layer. One factor that can affect anodization is the electrolyte solution, with previous research suggesting a measurable effect of supplemental acids added to a solution.

Background

The electrolyte solution used for anodization can have many different effects on the final product, including film thickness, film porosity, and corrosion potential, all of which affect the final corrosive physical resistance (Milani Hossein & Seyfinejad, 2022). Electrolyte solutions include various concentrations of acids such as sulfuric acid and phosphoric acid (Khan et al., 2019; Rashid, K., 2015). The electrolyte solution can be composed of just one acid or can be supplemented with an additional acid. Past research suggests that the addition of supplemental acids, such as acetic acid, citric acid, and tartaric acid, can have a positive effect on the oxide layer (Khan et al., 2019; Schneider, M. et al., 2024; Rashid, K., 2015).

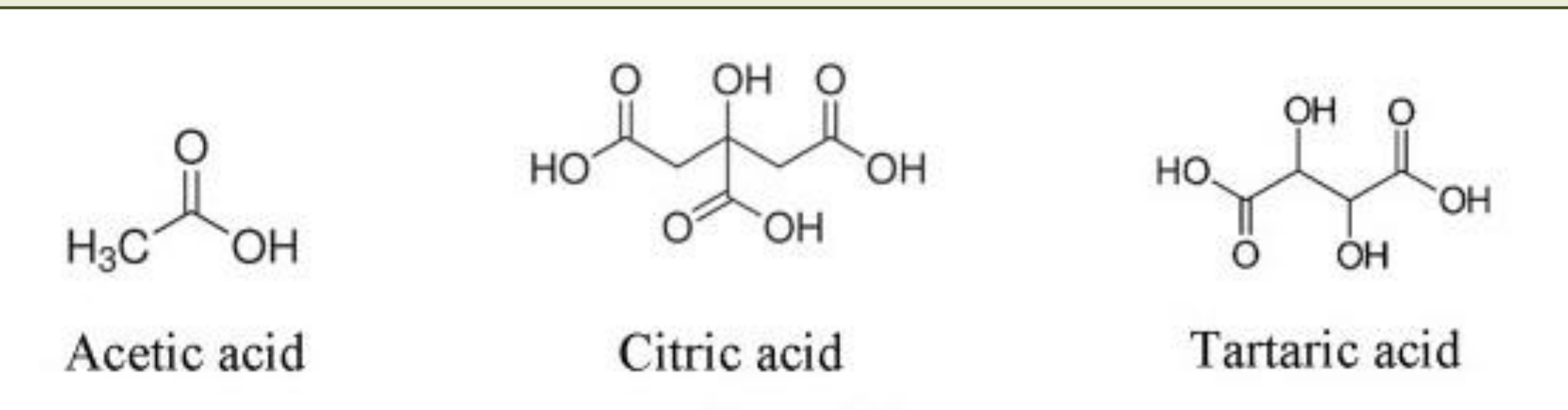


Figure 1. Chemical structures of acetic, citric, and tartaric acid, organic acids that are commonly found in foods and can be used as supplemental acids (Joye, I., 2019)

Purpose

Past research has not studied the effect of specific concentrations of supplemental acids when added to the electrolyte solution. Thus, this research aims to determine the optimal concentrations of supplemental acids to provide the best corrosive resistance. The hypothesis is that an increased amount of any supplemental acid increases the corrosive resistance of the aluminum.

Determining the optimal concentrations of supplemental acids when anodizing aluminum alloy 2024 to maximize corrosive resistance

Methodology

Preparation of aluminum

1. Aluminum alloy 2024 was sanded down.
2. Aluminum was degreased with acetone.
3. Aluminum was etched in 9g NaOH / 75mL water solution for 3 minutes.

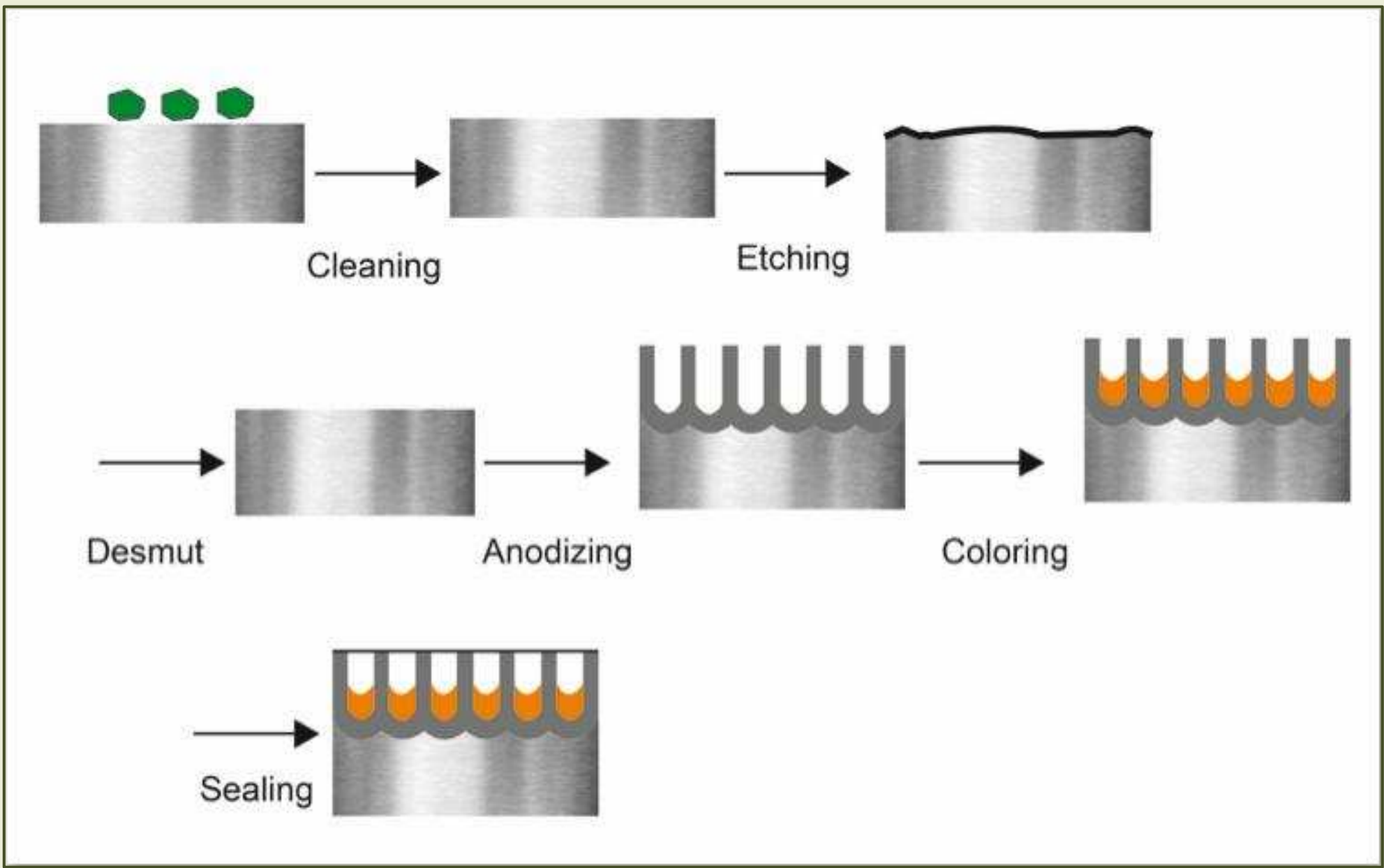


Figure 3. General overview of each step to anodize aluminum (Light Metal Age, 2021)

Analysis of Corrosive Resistance

1. Created testing solution of 3.5 % wt. NaCl.
2. Created setup for a potentiodynamic polarization test, where voltage applied is varied and the current passed is measured, with aluminum, silver chloride reference electrode, graphite counter electrode, and Rodeostat potentiostat.
3. The data collected from this setup was then analyzed using EC Lab, where values for corrosion potential and current could finally be obtained.

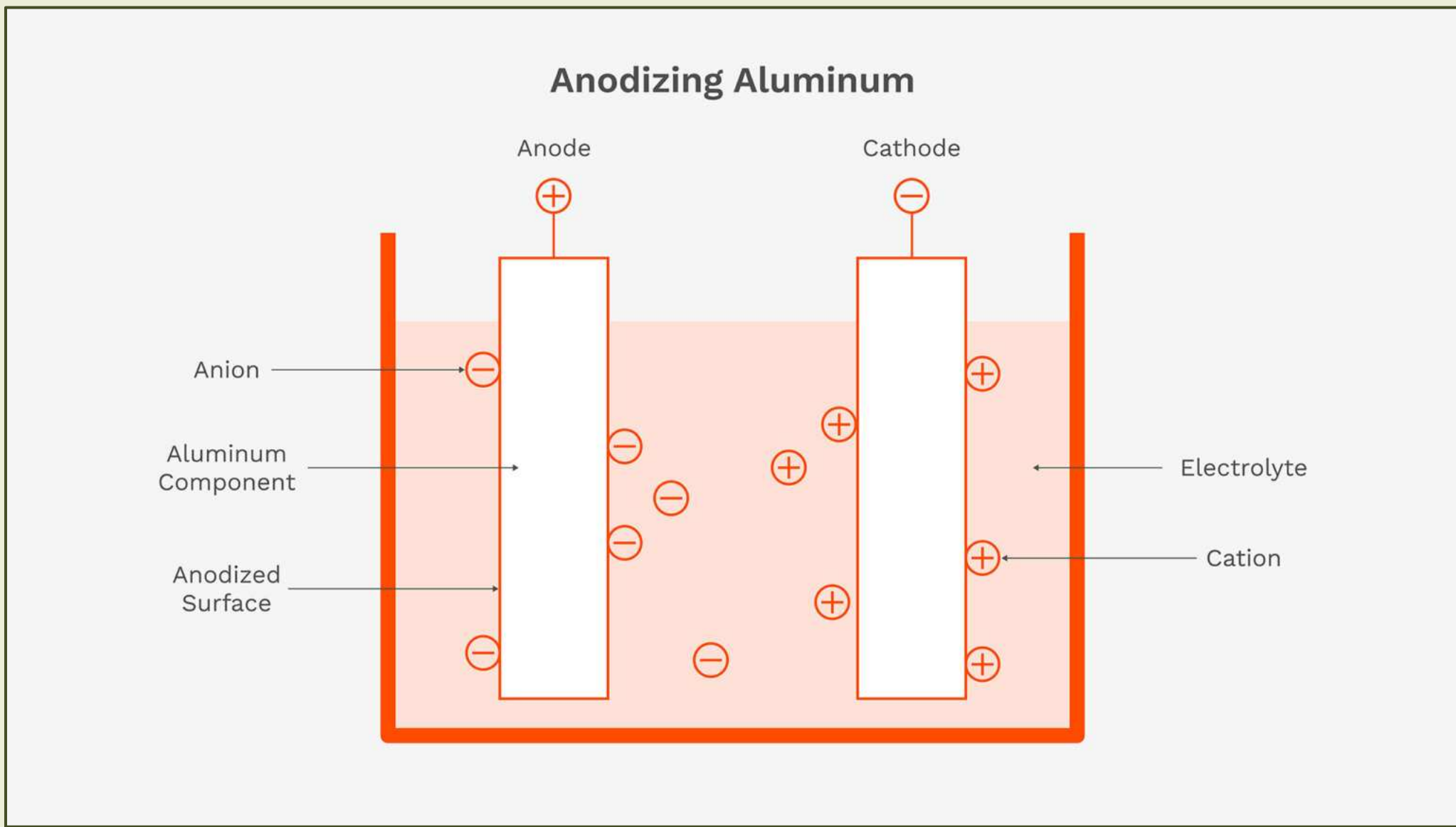


Figure 2. Labeled breakdown of the reactions during anodization (The Federal Group USA, 2024)

Anodization of aluminum

1. Electrolyte solution with 5 % sulfuric acid by weight and varying amount of supplemental acid was created (1, 3, 5 % wt. of acetic, citric, or tartaric acid, or 0 % wt. control).
2. Aluminum was anodized in the electrolyte solution at 25V for 30 minutes (with graphite counter electrode).
3. Aluminum was sealed by placing in 96 °C water for 20 minutes.

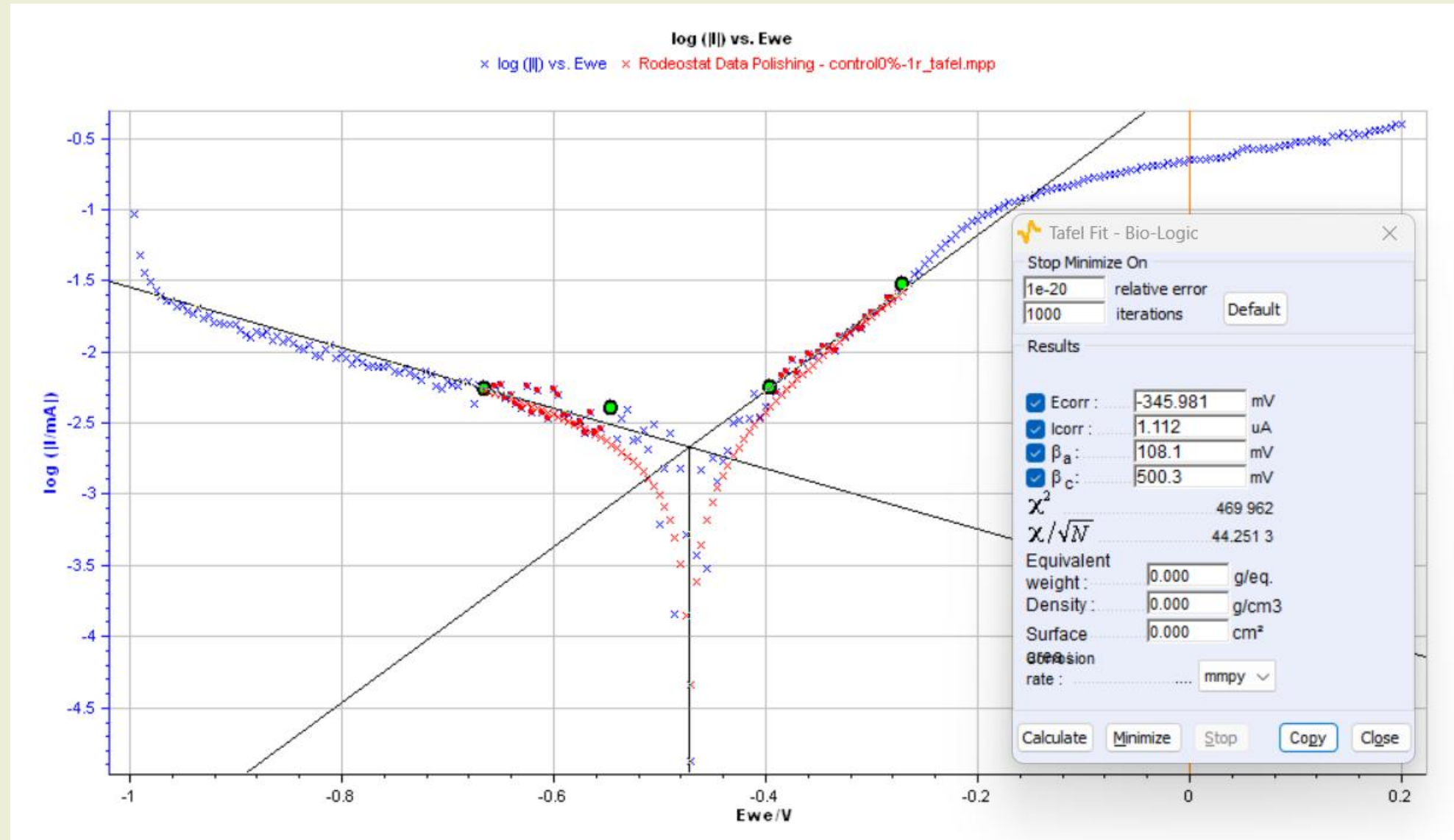


Figure 4. Tafel plot of analysis of the 5% acetic acid sample (Graph generated by the researcher using EC Lab)

Results

Analysis using Rodeostat and EC Lab demonstrated that **tartaric acid reached its most negative corrosion potential at 3 % wt., citric acid reached it at 5 % wt., and acetic acid remained around constant** with any concentration, decreasing slightly. Analysis also demonstrates that **for acetic and tartaric acid, as concentration increases, corrosion current increases**, and corrosion current for citric acid remains relatively constant regardless

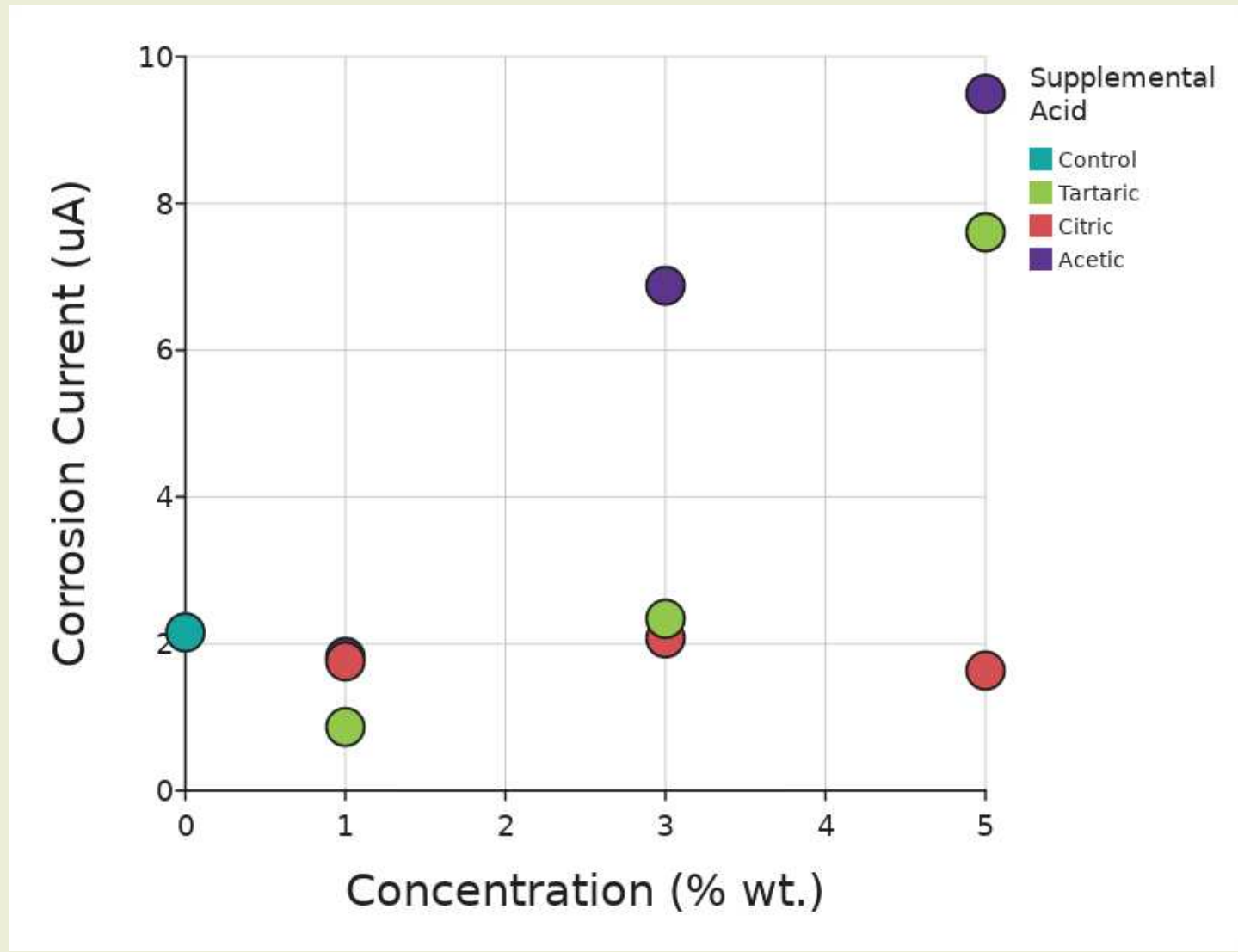


Figure 5. Graph of corrosion potential versus concentration of acid.

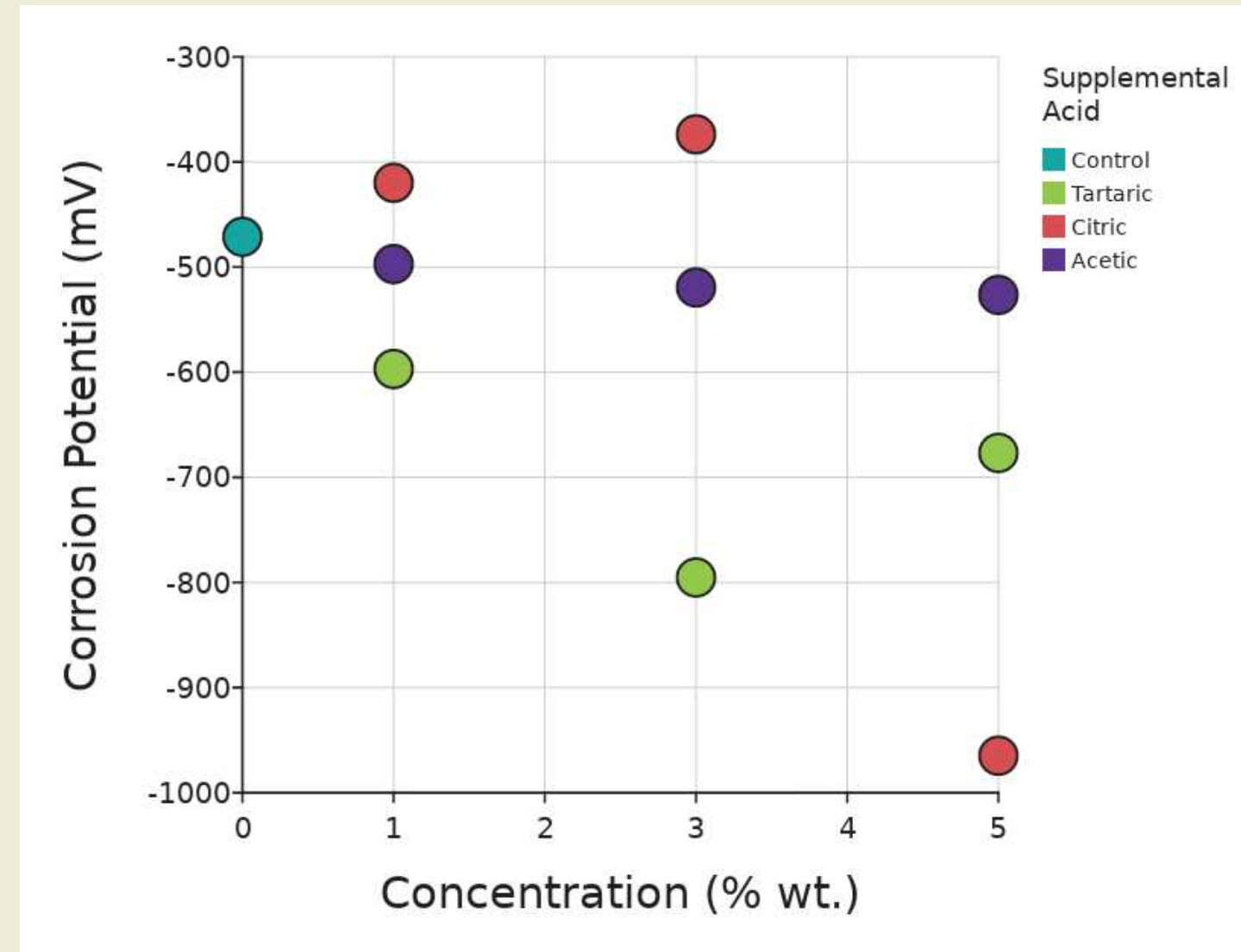


Figure 6. Graph of corrosion current versus concentration of acid.

The results of this research could be used to optimize the process of anodization by creating stronger anodized products with weaker concentrations of electrolyte solution. These experiments could be further expanded upon with morphological analysis of the aluminum itself using a scanning electron microscope.

References

Khan, M. F., Kumar, A. M., Ul-Hamid, A., & Al-Hems, L. M. (2019). Achieving non-adsorptive anodized film on Al-2024 alloy: Surface and electrochemical corrosion investigation. *Surfaces and Interfaces*, 15, 78–88. <https://doi.org/10.1016/j.surfin.2019.02.005>

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Rashid, K. (2015). Effect of Acetic Acid on Electrochemical Behavior of Sealed AA2319-T3 Al-Alloys Anodized in Phosphoric Acid Electrolytes. *Al-Khwarizmi Engineering Journal*, 11, 1–7.